

Business Central Custom API - Practice Problem

Interview-style AL development exercise. Build the solution yourself from the requirements below.

Scenario

A company uses Microsoft Dynamics 365 Business Central and also has an external web application used by employees to submit service requests. The company wants the external application to create, read, and update these requests in Business Central through a custom REST API.

Required AL Objects

- A custom table named **Service Request**.
- An enum for **Service Request Priority**.
- An enum for **Service Request Status**.
- A custom API page that exposes service requests.

Service Request Data

Field	Type / Requirement
Entry No.	Integer; automatically assigned
Customer No.	Code[20]; must reference an existing Customer
Title	Text[100]
Description	Text[250]
Priority	Enum: Low, Normal, High
Status	Enum: Open, In Progress, Closed
Created At	DateTime; automatically populated when created

Business Rules

- When a new service request is created, its status must automatically be set to **Open**.
- The **Created At** value must be set automatically when the record is created.
- The API must reject a request when the supplied Customer No. does not exist in the Customer table.
- External applications must not be allowed to manually set the internal record identifier.

API Requirements

Use the following API metadata:

Property	Required Value
Publisher	mycompany
Group	service
Version	v1.0
Entity Name	serviceRequest
Entity Set Name	serviceRequests

Required API Operations

- **GET** - Return all service requests.
- **GET by ID** - Return one service request.
- **POST** - Create a new service request.
- **PATCH** - Update an existing service request, such as changing its status.

Example Create Request

```
{
  "customerNo": "10000",
  "title": "Printer not working",
  "description": "The printer stopped working this morning.",
  "priority": "High"
}
```

Example Update Request

```
{
  "status": "In Progress"
}
```

Acceptance Criteria

- The extension publishes successfully to a Business Central environment.
- The custom endpoint is reachable through the Business Central API URL.
- GET requests return service request records.
- POST creates a valid service request and automatically assigns Status and Created At.
- PATCH can update an existing service request.
- POST with an invalid Customer No. fails with an appropriate Business Central error.